

Sterilization and inhibition of microbial activity in soil

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Abstract

This review examines methods to sterilize soil or inhibit microbial activity in soil. This is often necessary for experiments involving microbial survival, enzyme measurements, soil respiration, biodegradation or as controls in direct soil DNA extraction-PCR analysis.

Keywords: Bacteria; Chemicals; Fumigation; Soil; Sterilization

1. Introduction

The objective of soil sterilization is to destroy/eliminate microbial cells while, at the same time, not significantly altering the chemical and physical characteristics of the soil [1]. The sterilized soil is often used in estimates of soil biomass or as controls for soil enzyme assays, soil respiration, biodegradation of chemical(s) or for direct DNA extraction and polymerase chain reaction (PCR) analysis to verify the absence or presence of a specific DNA sequence [2]. In other experiments, chemicals, such as formaldehyde, and other commercially available disinfectants may be added to soil to determine its killing effect on genetically-engineered bacteria [3]. A concentration of 1% (v/w) formaldehyde (CH_2O) in 110 g of moist soil contained in polypropylene beakers at 0 and 7 days was effective in killing 10^9 *Pseudomonas aeruginosa* UG2Lr cells/g soil that were initially added to the soil [3]. Total indigenous,

aerobic heterotrophic bacterial colony forming units (CFU) decreased about 100-fold.

There are numerous difficulties associated with soil sterilization, especially if it is necessary to sterilize large volumes or masses of soil. The smaller the amount of soil, the easier it is to achieve complete sterilization. Moreover, sterilization requires destruction of both microbial cells and resistant spores which may germinate days, weeks or months after the sterilization process, if conditions are appropriate (e.g., nutrients, moisture, temperature) and the sterilization process was not initially complete.

It is necessary to verify that the soil samples subjected to the sterilization process are sterile. This is accomplished by preparing soil dilutions with known amounts of soil in sterile saline or sterile 0.1% (w/v) sodium pyrophosphate (adjusted to pH 7) and spread, plating the lowest dilutions (10^{-1} , 10^{-2}) on nutrient agar or another general plating medium (tryptone-yeast extract agar) for enumeration of colony forming units (CFU). This method will provide an estimate of the viable heterotrophic bacteria per g dry soil. If CFU are present after

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incubation at room temperature or 28°C for 1 or more days, then the soil samples must be sterilized again or discarded. It is advisable to incubate the plates for 2–10 days to ensure that slow growing or injured microbial cells are given the opportunity to form colonies if present. Longer incubation times of 2–3 weeks and a medium with a low nutrient level may be needed to permit growth of oligotrophic soil bacteria. Alternatively, 1 g samples of sterilized soil are aseptically placed into 10 ml of sterile nutrient broth, incubated at room temperature or 28°C and observed for turbidity (microbial growth) for the next 7–10 days [1]. If turbidity occurs in the broth, the soil is not sterile.

2. Soil sterilization

Table 1 summarizes methods and provides references for soil sterilization. The article by Tuominen et al. [4] on methods for inhibiting bacterial activity in sediment is also recommended. The methods used to treat the sediment included formaldehyde, HgCl_2 , autoclaving and γ -radiation. Dilute acid-free formaldehyde (0.02–0.04%) buffered with CaCO_3 was the most promising for inhibiting bacterial activity (measured by incorporation of ^3H thymidine into DNA) in sediment. In addition, dilute formaldehyde did not cause changes in the dissolved nutrient concentrations in the sediment or decrease sediment pH.

The reader is also referred to the excellent refer-

ence text by Block [5] on disinfection, sterilization and preservation. Although this text does not deal specifically with soil, it is a complete up-to-date reference text for disinfection, sterilization and preservation. In numerous soil experiments the objective is to sterilize the soil so it is devoid of any living organisms and biological activity. However, in some experiments disinfectant agents (usually chemicals or physical agents such as ultraviolet light or X-rays) are used to destroy disease causing or other harmful microorganisms. Germicides are agents that destroy microorganisms, especially pathogenic microorganisms [5]. For a complete list of definitions, the reader is referred to Block [5]. With respect to soil sterilization, repeated autoclaving and irradiation are the most widely used and efficient methods to achieve this task. The method used by the researcher will depend on the experimental design and the length of time the soil must be maintained sterile after the initial sterilization. If the method causes changes in soil pH or the nutrient status of the soil, erroneous interpretations of the data could creep into the research.

3. Autoclaving (moist heat)

Autoclaving is widely used to sterilize soil samples as this equipment is readily available in most research and teaching laboratories. Soil is autoclaved at 121°C at 1.1 atm (approx. 16 lbs/in²; 1.137 kg/cm²) for a minimum of 20–30 min, but generally

Table 1
Methods of soil sterilization

Methods	References
Autoclaving (moist heat)	[1], [4], [6], [7], [13], [21], [25]
Dry heat	[1], [21]
γ -Irradiation	[1], [4], [6], [10], [11–13], [21], [24], [26]
Microwaves	[1], [14–17], [21]
Gaseous chemicals	
Chloroform	[6], [18], [19], [21]
Ethylene oxide	[1], [6], [22–24]
Propylene oxide	[1], [20], [21], [25]
Methyl bromide	[1], [13], [26]
Mercuric chloride	[1], [21]
Sodium or potassium azide	[1], [6], [25]
Formaldehyde	[3], [33]

Note, reference 4 deals with sediment.

longer (1 h). The length of time should be increased when larger amounts of soil are used. For example, Alef and Nannipieri [6] recommended 1 h of autoclaving is required for soil samples larger than 500 g. The soil can be autoclaved in glass beakers, glass Petri dishes (lids ajar) vials or stainless steel trays with lids ajar. The containers can be covered with aluminum foil. The soil should be spread out in a thin layer (1–2 cm) to allow for penetration of steam during the autoclave cycle. Beakers, stainless steel autoclave trays or polypropylene trays completely filled with soil is difficult, if not impossible to completely sterilize even after 2–3 repeated autoclavings. The soil should not be packed or compressed into containers, but left uncompressed to allow for penetration of the steam. In addition, the soil should not be saturated with water if this is possible. Air-dried soil or soil with a low water content (less than about 60% of moisture holding capacity) will allow for better sterilization. It is also noteworthy that autoclaving destroys soil structure and releases ammonium-N and amino acids [6]. It has also been reported that autoclaving can decrease the surface areas of clays [7] and alter the surface charge of sandstone [7].

Autoclaved soil samples have been used to study survival and respiratory activity of genetically-engineered *Pseudomonas* spp. [8]. Bacteria introduced into sterile soils were enumerated to determine their survival in the absence of competing indigenous soil organisms and to remove protozoans from the soil which ingest soil bacteria. Autoclaved soil has also been inoculated with a known number of bacterial cells and then extracted to determine the DNA content of the inoculated sterile soil [9].

4. Dry heat

Dry heat sterilization is achieved by heating soil samples at 200°C for a minimum of 24 h [6]. Heating 50 g amounts (in 15 cm glass Petri plates with lids partly offset) of soil (silty clay loam) at 160°C for 3 h is not sufficient to sterilize soil. It is important for the soil depth to be only a few cm. For example, a large glass Petri plate (15 cm diameter) where the lid can be replaced after sterilization is an excellent container. The cover should be replaced

before removal from the oven to prevent recontamination of the soil. Alternatively, soil can be placed a few cm deep in glass beakers, glass vials, flasks or stainless steel trays with lids that can be replaced after sterilization. Aluminum foil can also be used to cover containers. It is noteworthy that bacterial spores are more resistant to dry heat sterilization than autoclaving. For this reason it may be necessary to wet the soil samples (if necessary) and incubate the soil for 1–2 days to allow spores to germinate prior to dry heat sterilization. A sterility check as described earlier is necessary to ensure that the sterilization process was successful.

5. γ -Irradiation

The main advantage of irradiation is that minimal disturbance of soil occurs during this process [6]. However, γ -irradiation using a ^{60}Co source can only be done at an irradiation facility. Soil adjusted to the desired water content for experiments are placed in sealed polyethylene sacs and irradiated at 2.5 Mrad at a rate of 2 Mrad/h [6]. After irradiation, the soils are maintained under sterile conditions to prevent subsequent contamination by non-aseptic handling, exposure to the open atmosphere or use of non-sterile water or chemicals added to the soil. It was recommended by Alef and Nannipieri [6] that the soil samples be maintained frozen before and after irradiation. This decreases changes in the soils prior to experimentation. γ -irradiation creates free hydrogen and hydroxyl radicals that are reactive as reducing and oxidizing agents and cleave C-C bonds [4]. In addition, radiation can cause depolymerization of carbohydrates like cellulose [4].

Soils can also be irradiated at a higher dose. For example, 4 Mrad has been used to sterilize soils from Canada and The Netherlands for use in studying survival of *Pseudomonas fluorescens* cells introduced into the sterile soils [10]. Serial dilutions (10^{-1}) of the irradiated soil plated onto tryptone-yeast extract agar revealed no colony forming units (CFU) were present. Jackson et al. [11] irradiated 30 g soil samples in plastic culture dishes with γ -irradiation ranging from 0 to 3 Mrad (1 rad = 100 erg/g). One Mrad was required to kill all fungi, while 2–3 Mrads were necessary to eliminate all

bacteria. Generally, the larger the microbial cells, the less ionizing radiation is required to kill the cells [12]. In addition, microbial cells are destroyed by ionizing radiation in an exponential manner [12]. The reader is referred to McLaren [12] for a history on the use of radiation as a technique in soil biology/biochemistry.

Eno et al. [13] reported that the availability of N, P and S increased in some soils exposed to radiation. McLaren [12] also reported that irradiation can release manganese, ammonium, soluble C, organic N and phosphorous from the soil microorganisms and to a lesser amount from soil humus.

6. Microwaves

Two-thousand-four-hundred-and-fifty MHz microwave radiation has been experimented with as a method to sterilize soil. Microwaves cause water in the soil and cells to absorb heat which results in high temperatures that kill the soil organisms. For this reason, the lethal action of microwaves is higher in moist soils than dry soils [6]. Vela and Wu [14] reported that bacteria, actinomycetes, fungi and bacteriophages exposed to 2450 MHz microwaves were only destroyed when water was present. Lyophilized organisms were not destroyed by microwaves as there was no water present to absorb adequate energy to raise the temperature and bring about killing. Wainwright et al. [15] demonstrated that the microwave treatment of soil destroyed fungi but not heterotrophic bacteria. Microwaves are rarely, if ever used to treat soils in microbiological research.

Microwaves (625 watts, 2450 MHz) have been used to reduce soilborne plant pathogens [16]. For example, soil containing plant pathogens placed in plastic sacs was exposed to microwaves. One kg of soil exposed to microwaves for 150 s eliminated *Pythium*, *Fusarium* and all nematodes, except *Heterodera glycines* [16]. Four kg of soil microwaved for 425 s yielded similar results [16]. Polypropylene sacs were more suitable than polyethylene sacs which were reported to break. Speir et al. [17] observed that soil fungi are more easily destroyed than soil bacteria by microwave treatment. They also reported that a 90 s microwave treatment eliminated

microbial biomass equivalent to that achieved by CHCl_3 fumigation. The researchers used 25 g of moist soil in plastic Petri plates with the lids on. A 600 W, 2250 MHz oven was used. A 10 s treatment of 25 g of soil, raised the soil temperature 13°C [17]. A 30 s treatment caused the soil temperature to increase 34°C [17].

7. Gaseous chemicals

7.1. Chloroform

Chloroform (CHCl_3) is highly volatile and has been used to fumigate soils for the estimation of microbial biomass [18]. A description of this method has been provided by Alef and Nannipieri [6]. Briefly, 250 g (or less) soil in glass beakers are placed into a desiccator lined with moist paper containing a beaker of alcohol-free chloroform and anti-bumping chips [6]. Upon evacuation of the desiccator, the chloroform boils. Evacuation is halted and the desiccator maintained for 24 h at which time the beakers containing soil or chloroform and the paper are removed under a chemical hood. Repeated 3 min evacuations of the desiccator will remove chloroform from the soil [6]. It is recommended that a total of 8 evacuations be used to remove the chloroform. To ensure that an explosion hazard is not created, the first 3 evacuations should be done using a water aspirated vacuum. The remaining evacuations are done with an oil vacuum pump. Always, ensure there is no exposure to chloroform or that an explosion hazard is created.

Jenkinson and Powlson [19] reported that chloroform fumigation of soil caused an immediate increase in ammonium and organic C extractable with 1 N K_2SO_4 .

7.2. Ethylene oxide, propylene oxide, methyl bromide (monobromomethane)

Ethylene oxide ($\text{C}_2\text{H}_4\text{O}$) is a colourless, flammable gas at room temperature and pressure. Propylene oxide ($\text{C}_3\text{H}_6\text{O}$) is a colourless, extremely flammable liquid that can alkylate functional groups of proteins [20]. Methyl bromide (CH_3Br) is a colourless gas. Commercially available sterilization

units are used with ethylene and propylene oxide [1]. Soil is moistened and placed in a flask, vial or test tube and incubated for 1–2 days to permit germination of bacterial spores. The soil is then sterilized using an application rate of 10 mol ethylene oxide for 24 h or 7 mol propylene oxide per 1 kg soil for 48 h [1]. For 100 g (or smaller soil samples), a vacuum desiccator can be used in a properly vented fume hood [1]. The boiling point of ethylene oxide is 11°C, and propylene oxide 35°C [1]. Therefore, ethylene oxide must be kept cold, and added to the vacuum desiccator which is quickly sealed and the vacuum established without an explosion hazard. The fumigant is removed from the soil samples by a minimum of 6 evacuations [1].

It is noteworthy that ethylene and propylene oxide can increase the soil pH during fumigation due to esterification of carboxyl groups in soil organic matter [1]. Wolf et al. [21] reported that soil pH was increased after treatment with propylene oxide. A similar observation was reported by Clark [22] where the soil pH increased from 6.15 to as high as 8.14 depending on the soil and the total hours the soil was exposed to the compound. The longer the exposure time, the higher was the increase in soil pH. Allison [23] also described a vapor-phase apparatus for sterilizing soil with ethylene oxide. An increase in soil pH was also observed in ethylene oxide treated soils. Ethylene oxide has been reported to be transformed to ethylene glycol and polyethylene glycol during soil sterilization [1]. Griffiths [24] reported that the growth of *Pseudomonas fluorescens*, the protozoan *Colpoda steinii* and the nematode, *Rhabditis sp* were inhibited in soil previously treated with ethylene oxide. It was proposed that residues of ethylene chlorohydrin and ethylene glycol were present in the soil from the sterilization process when using ethylene oxide.

Twenty-five g portions of soil treated with 1 or 2 ml of propylene oxide for 2 days at 27°C was effective as a soil sterilant [25]. Following this, the soil contained in flasks was allowed to off gas for 2 days in a laboratory exhaust hood. However, growth of alfalfa and wheat were retarded in the treated soil [25]. Bartlett and Zelazny [20] have described a method for the sterilization of small amounts (100 g dry soil) of soil in polyethylene bags sealed with rubber bands and placed in a 14-l aluminum pressure

cooker. Propylene oxide was added at 320 ml for each 1.6 kg dry soil. The pressure cooker was sealed and placed in a 60°C water bath for 15 min in a fume hood [20]. The propylene oxide boils at 35°C and diffuses into the polyethylene sacs containing the soil. The pressure cooker was then cooled to 20–22°C in a water bath and a screw clamp on the outlet slowly open (prevent sac containing soil from breaking) under a proper fume hood to allow the gas to escape. Additional steps are also required in a vacuum oven under 0.8 atm for 3 days to remove residual propylene oxide. The toxic exhaust from the vacuum oven can be collected in a water trap [20].

Methyl bromide (Brom-O-Gas, Great Lakes Chemical Corporation, West Lafayette, Indiana, USA) is an extremely poisonous gas (boiling point 5°C) that has been used to sterilize soil [1]. A properly vented chemical fume hood is required. Soil samples can be placed in polyethylene bags in a desiccator and fumigated for 24 h. The Brom-O-Gas contains 2% chloropicrin (tear gas) as an easily detectable warning gas [1]. Because of the flammable nature of methyl bromide, commercially available preparations also contains carbon dioxide or dichlorodifluoromethane [1].

Methyl bromide decomposes to form bromide and methanol. Therefore, elevated bromide concentrations in the soil may occur. Methyl bromide has been used to fumigate soil to destroy vesicular-arbuscular mycorrhizal (VAM) fungi [26].

In addition, methyl bromide is being banned for use as a soil sterilant due to its adverse effect on the ozone layer.

8. Chemicals

8.1. Mercuric chloride

Mercuric chloride (HgCl_2) can be used to inhibit soil microbial activity by combining with cellular proteins containing sulfhydryl groups [2]. Generally, a concentration of 500 mg HgCl_2 /kg soil is satisfactory [4]. A stock solution of the chemical can be prepared and added to soil which can be contained in a variety of glass, polypropylene or stainless steel containers. It is noteworthy that mercury can bind to glass. The containers should be covered and placed

in a chemical fume hood in the event soil microorganisms are present that are capable of enzymatically volatilizing Hg^{2+} to volatile Hg^0 (via the mercuric reductase enzyme) [27–30] and methylating mercury under anaerobic conditions [31]. Because of the potential for these biological reactions, the fact that mercury treated soil poses a disposal problem and plants can not be grown in the soil, the usefulness of mercury as a soil inhibitor is decreasing.

However, if the soil does not need to be used for growth of plants, mercuric chloride is a suitable inhibitor for preventing microbial activity in aerobic soils. In addition, its use results in minimal changes in soil chemical and physical properties.

8.2. Sodium azide

A stock solution of sodium azide (NaN_3) can be added to soil samples contained in glass beakers, flasks and vials that can later be sealed. The final concentration of sodium azide is usually between 1 and 10% [6]. Beakers can be covered with aluminum foil while beakers and flasks can be covered with foil or sealed with sterile rubber septa or stoppers. The latter method prevents gas exchange and the soils system may become anaerobic depending on the length of incubation and water content of the soil. The soil samples are not disturbed unless the soil is mixed after addition of the solution. Sodium azide is toxic and MSDS information must be consulted for safe handling of the sodium or potassium azide and disposal of the treated soil. It is also noteworthy that a high sodium azide concentration (10%) removes sodium ions from soil in subsequent leaching experiments [6]. If azide is used to inhibit soil respiration, it is necessary to poison the soil several hours prior to commencing the soil respiration experiment. Sodium azide is an inhibitor, not a soil sterilant. It has also been reported that sodium or potassium azide (KN_3) is not suitable for inhibiting soil respiration as it can cause errors in carbon dioxide evolution in soil as well as increasing soil pH [32]. The increase in pH will be a function of the soil buffering capacity. In soil, azide can be transformed to hydrazoic acid [32].

9. Conclusions

Numerous methods have been used to sterilize soil or inhibit microbial activities in soil. The method used often will depend on the facilities available and the design of the particular experiment. It is imperative, after the sterilization process has been completed, to verify that the soil microbial population was eliminated. Secondly, the sterilized soil may become contaminated after the sterilization process. It is necessary to know if this happens, especially in experiments where the soil must be maintained sterile for weeks, months or longer.

Sterilization usually alters the chemical and physical properties of soil in addition to eliminating organisms. The method used will depend on the research being conducted. This is especially important when a chemical is used to destroy soil microorganisms and the chemical residue may later interfere with subsequent measurements in the soil. Currently, many laboratories use repeated autoclaving or γ -irradiation to sterilize soil samples. These methods also have the advantage of not creating a chemically contaminated soil that is difficult to dispose of. Autoclaving is also inexpensive in that all microbiology laboratories will have one or more autoclave facilities. Therefore, no additional expenses are necessary.

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